

PROJECT TITLE

AUTOMATED MULTIMODAL 3D BRAIN TUMOR ANALYSIS USING FEDERATED LEARNING AND PROGRESSION FORECASTING

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PROBLEM STATEMENT

A Federated Multimodal AI Framework with QPSO for Brain Tumor Classification, 3D Glioma Segmentation, and Tumor Progression Analysis

INTRODUCTION

Brain tumours require highly accurate detection, segmentation, and continuous monitoring to support timely clinical decisions. Traditional manual analysis of MRI scans is slow, subjective, and limited by human fatigue. Existing AI systems mostly perform isolated tasks such as basic classification or 2D segmentation without understanding tumour evolution over time. This project aims to build an integrated, clinically oriented AI pipeline capable of 3D multimodal tumour segmentation, tumour classification, longitudinal tumour progression (time-travel) analysis, and federated multi-hospital learning. The system is motivated by the need for precision, explainability, scalability, and privacy-preserving collaboration in neurodiagnostics.

ABSTRACT

This project proposes a multimodal AI framework for brain tumour diagnosis and monitoring. The system integrates **3D Attention-U-Net based segmentation, federated tumour classification, and longitudinal tumour progression forecasting**. Federated learning enables multiple hospitals to collaboratively train models while keeping patient data private. A **Quantum Particle Swarm Optimization (QPSO)** based aggregation strategy improves federated learning convergence. The system also predicts future tumour growth from longitudinal MRI scans using mathematical models and **LSTM** networks. The integrated pipeline provides **segmentation maps, tumour classification, growth prediction, and clinical risk assessment**, enabling intelligent clinical decision support.

OBJECTIVES

- Develop 3D multimodal tumor segmentation model
- Implement **federated** tumor classification across hospitals
- Predict **tumor progression** using time-series models
- Integrate **segmentation + classification + forecasting**

LITERATURE SURVEY

S.No	Author & Year	Paper / Study	Methodology	Architecture Used	Research Gap
1	Edla Divyansh Teja & Indhumathi, 2025	Enhancing Federated Learning with Quantum-Inspired Particle Swarm Optimization: An IID MNIST Study	MNIST dataset partitioned across 5 IID clients; each client trains locally for 10 epochs using Adam optimizer; QPSO applied as layer-wise aggregation using quantum-inspired probabilistic position updates; compared against FedAvg using accuracy, loss, confusion matrix, and per-class performance	CNN with 32-filter conv layer, max-pooling, 64-unit FC layer, softmax output; FedAvg and QPSO aggregation	Study restricted to IID data distribution only; real-world FL scenarios involve non-IID and heterogeneous data across clients; no comparison with proximal-term methods like FedProx; QPSO computational overhead across larger client pools remains unevaluated
2	Pati et al., 2022	Federated Learning Enables Big Data for Rare Cancer Boundary Detection	Largest real-world FL study for brain tumor segmentation across 71 institutions, 6 continents, using 6,314 glioblastoma patients (25,256 MRI scans)	Standard FedAvg with institutional consensus training; U-Net based segmentation	Uses standard FedAvg (no QPSO or fairness-aware aggregation); focuses only on segmentation; no per-institution fairness analysis; no Non-IID experimental setups — institutions contribute naturally but fairness gap isn't measured
3	Isensee et al., 2021	nnU-Net: A Self-Configuring Method for Deep Learning-Based Biomedical Image Segmentation	Fully automated pipeline that self-configures preprocessing, architecture, training, and post-processing based on dataset properties; evaluated on BraTS 2021	Self-configuring 3D U-Net (nnU-Net)	No attention mechanism — relies purely on convolutions, missing long-range spatial dependencies; does not leverage multi-modal attention gating for selective feature focus on tumor sub-regions
4	Subramanian et al., 2021	Spatio-Temporal ConvLSTMs for Tumor Growth Prediction by Learning 4D Longitudinal Patient Data	ST-ConvLSTM learns inter-slice 3D contexts and temporal dynamics from 4D longitudinal MRI to predict future tumor volumes and cell density	ST-ConvLSTM (Spatio-Temporal Convolutional LSTM)	Small dataset (33 patients); no hybrid mathematical + DL approach — purely data-driven; no biological growth models (Gompertz, Logistic) for interpretability
5	Zhang et al., 2023	Self-Supervised Temporal Deep Learning for Pediatric Glioma Recurrence Prediction	Self-supervised temporal learning on serial MRI scans; encodes longitudinal spatial features via ResNet and models temporal dynamics for 1-year recurrence prediction	ResNet encoder + Temporal self-supervised framework	Focused on binary recurrence prediction (yes/no), not continuous volume trajectory forecasting; no mathematical growth curve fitting; no hybrid Math + LSTM approach for interpretable kinetics modeling

PROPOSED SYSTEM

01

3D Brain Tumor Segmentation

- 3D Attention U-Net (5.9M params) on BraTS 2021, 4 MRI modalities → 3 tumor regions
- Live inference pipeline with MONAI transforms and sliding window
- Streamlit app: Slice-by-slice viewer + interactive 3D Plotly rendering
- Side-by-side Ground Truth vs AI Prediction comparison

02

Federated Learning — Brain Tumor Classification

- FedAvg, FedProx, and novel Layer-by-Layer QPSO across 3 simulated hospitals
- SimpleCNN (~120K params), 4-class classification, 100 FL rounds per setup
- QPSO achieves 3.5× better fairness than FedAvg with <1% accuracy trade-off
- Streamlit demo with multi-model comparison and ensemble voting

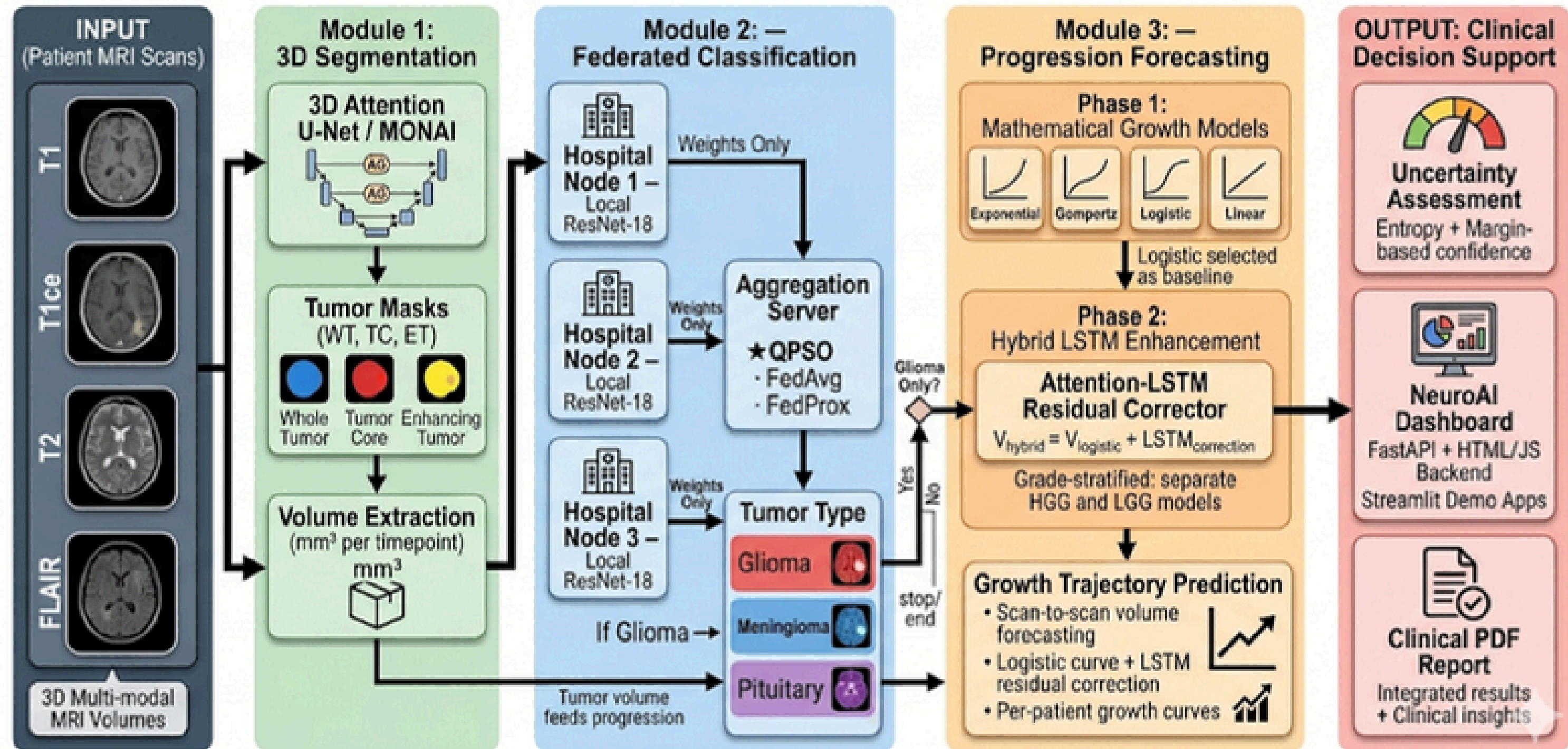
03

Tumor Progression Analysis

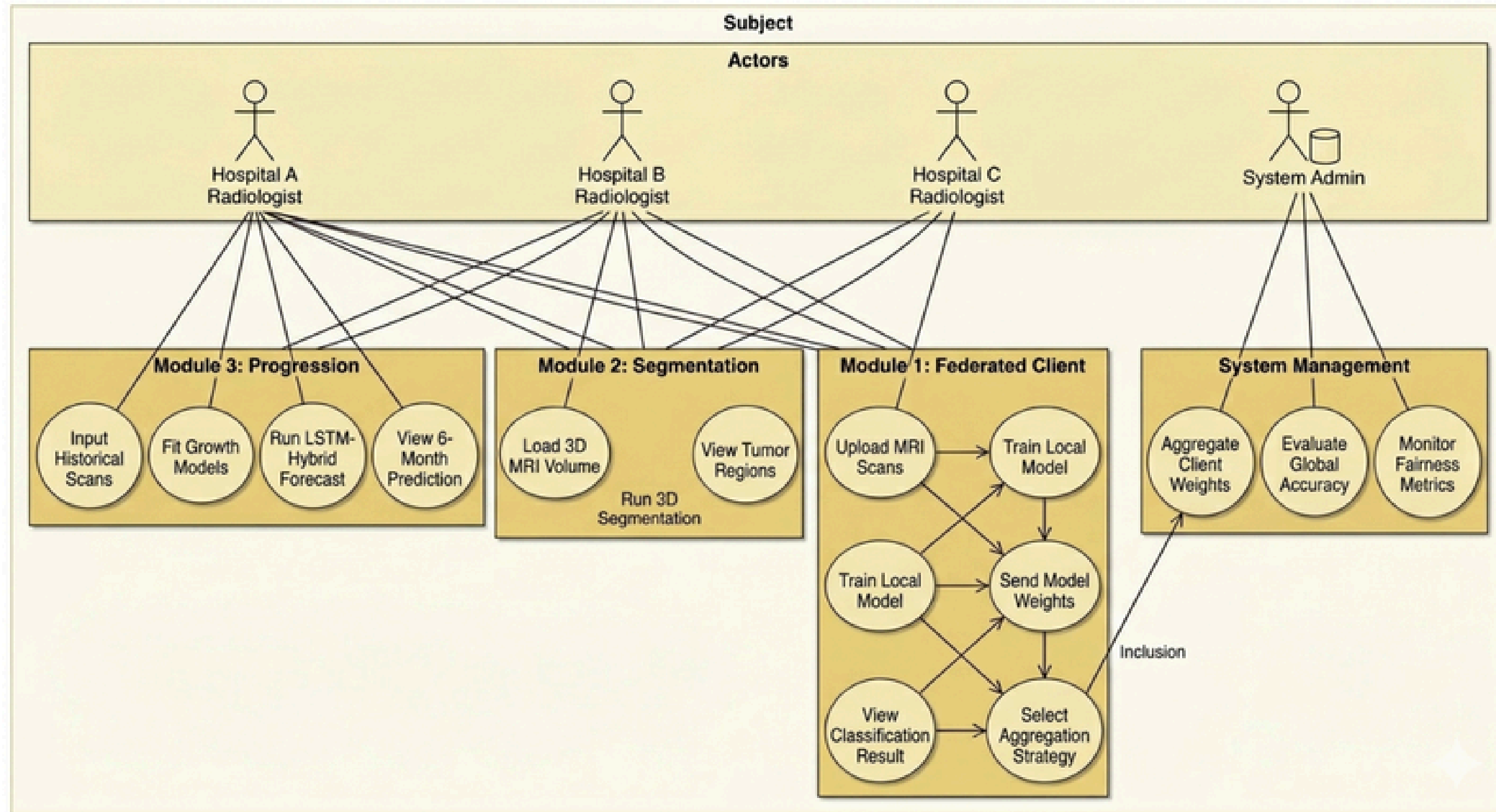
- Track tumor volume changes across longitudinal MRI scans using segmentation outputs
- Estimate growth rates and predict trajectory for clinical decision support
- Goal: early detection of aggressive growth patterns to guide treatment planning

SYSTEM ARCHITECTURE

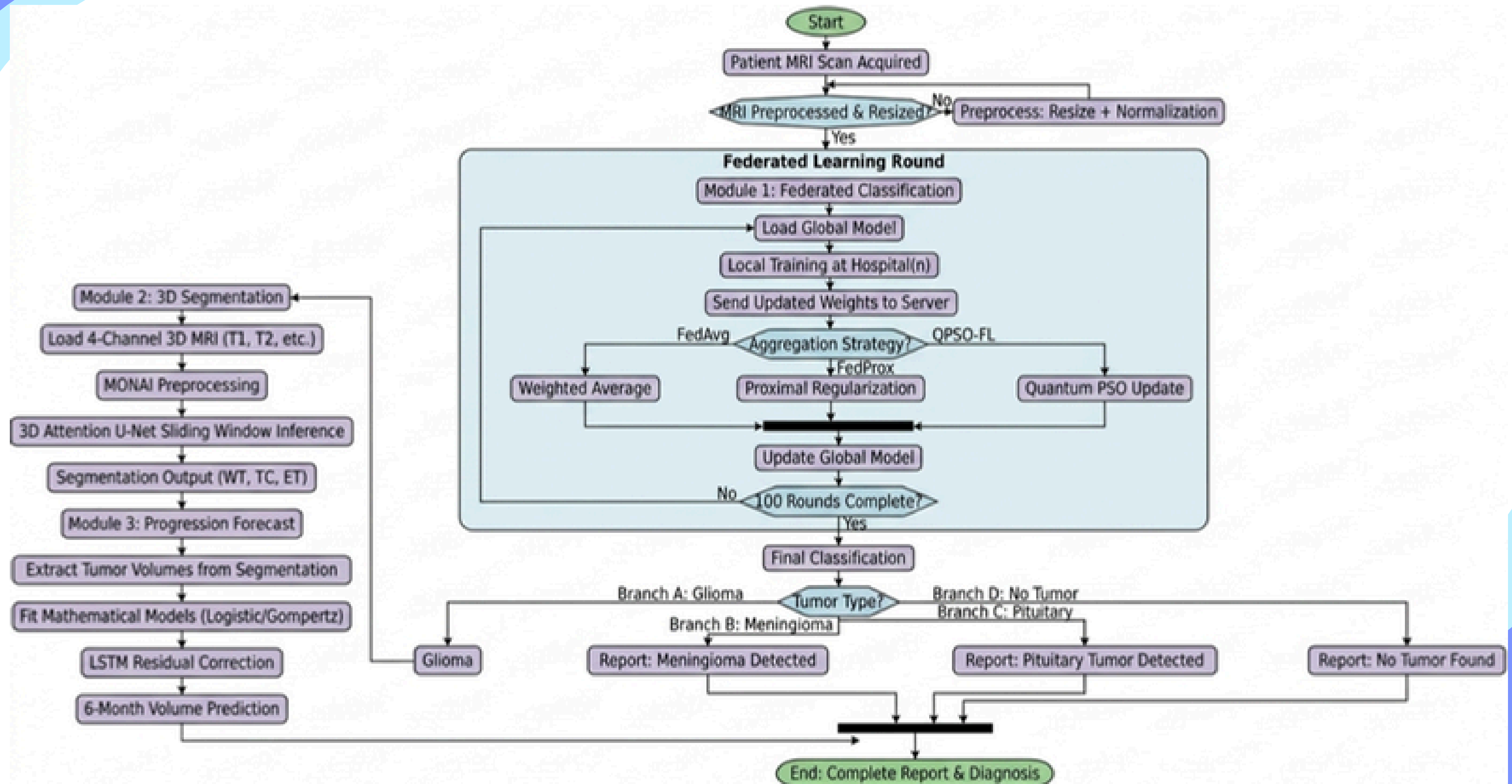
END-TO-END BRAIN TUMOR MANAGEMENT PIPELINE



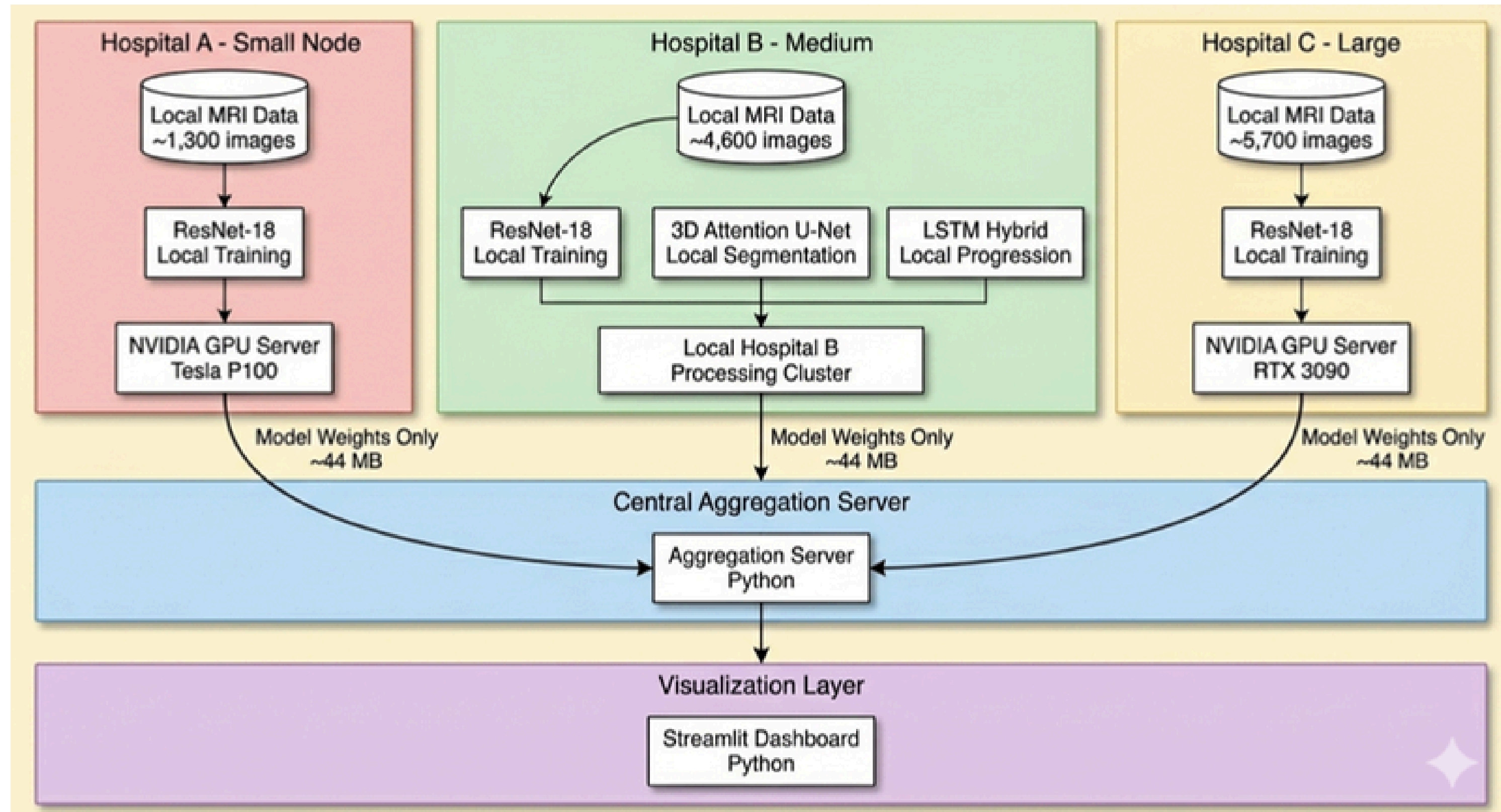
USE CASE DIAGRAM



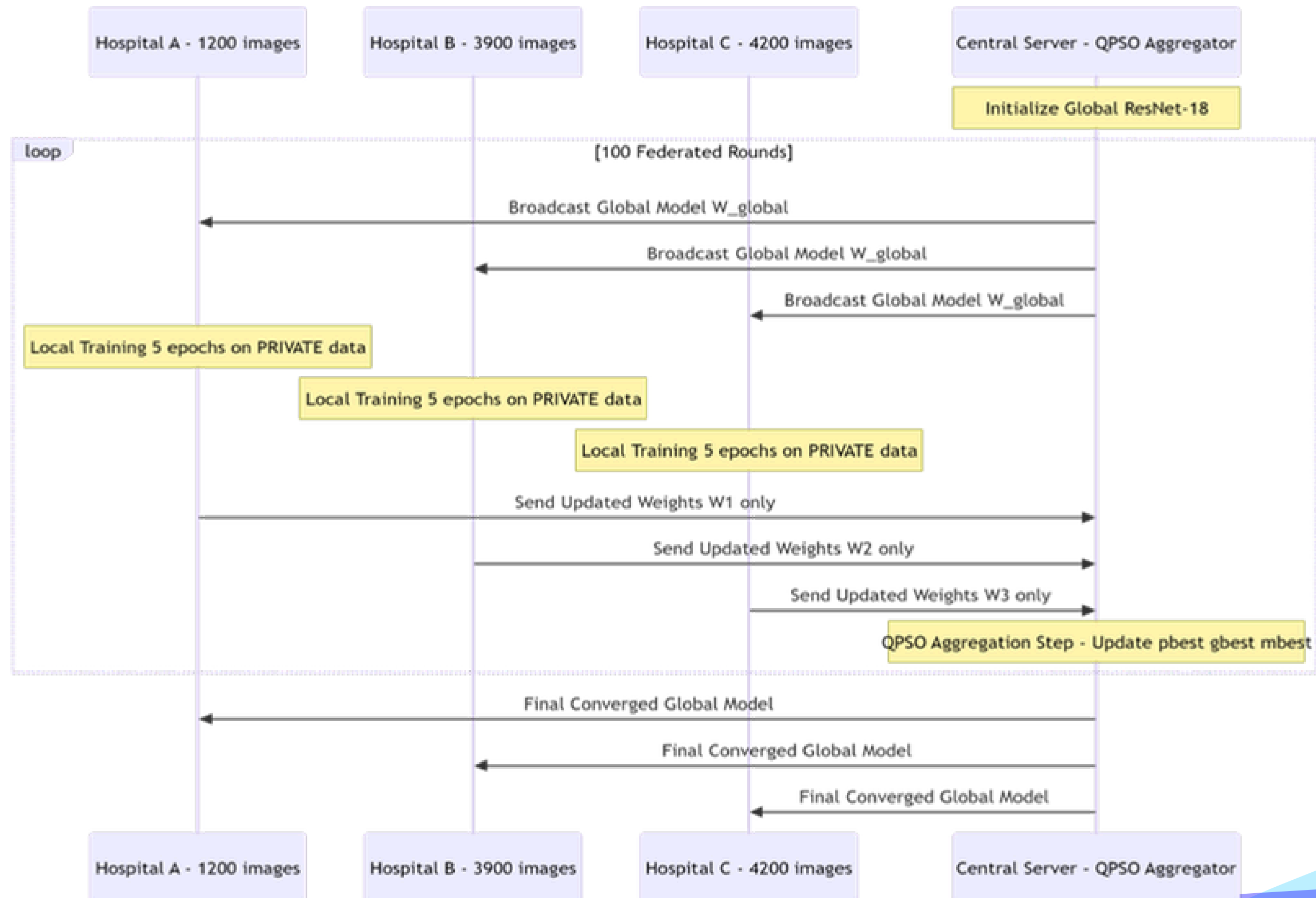
ACTIVITY DIAGRAM



DEPLOYMENT DIAGRAM



FL - SEQUENCE DIAGRAM



DATASETS

Property	Classification (FL)	Segmentation	Progression
Dataset	Masoud (Kaggle) + BRISC 2025	BraTS 2021	MU-Glioma-Post (TCIA)
Total Images / Patients	~9,300 MRI images	1,251 patients	203 patients
Data Type	2D MRI slices (224×224)	3D volumes (240×240×155)	Longitudinal 3D MRI series
Modalities	Single-channel MRI	T1, T1ce, T2, FLAIR	Multi-timepoint volumes
Classes / Labels	Glioma, Meningioma, Pituitary	WT, TC, ET masks	791 timepoints across patients
Format	JPEG/PNG	NIfTI (.nii.gz)	NIfTI (.nii.gz)
Split	Distributed across 3 clients	Train/Val/Test	155 valid trajectories extracted

CLASSIFICATION MODULE

METHODOLOGY:

CNN-based Classification using ResNet-18 (ImageNet pretrained, 11.2M params)

Federated Learning: FedAvg, FedProx, FedQPSO (custom layer-wise aggregation)

Quantum-Inspired Optimization: QPSO with cross-client validation fitness

Non-IID Data Handling: Natural heterogeneity + Label skew (70/10/10/10)

Evaluation: Per-client fairness analysis, ROC-AUC, Confusion Matrices

TECHNOLOGIES USED:

- PyTorch 2.x for model training & federated simulation
- torchvision for ResNet-18 & image transforms
- PIL / OpenCV for MRI preprocessing (resize, normalize)
- Matplotlib / Seaborn for visualization (ROC curves, fairness plots)
- NumPy / SciPy for statistical significance testing (paired t-test, Cohen's d)
- Kaggle Tesla P100-16GB for GPU-accelerated training

SEGMENTATION MODULE

METHODOLOGY:

3D Volumetric Segmentation using Attention U-Net with Attention Gates

Multi-modal MRI Input: 4 channels (T1, T1ce, T2, FLAIR)

Multi-class Output: Whole Tumor (WT), Tumor Core (TC), Enhancing Tumor (ET)

Loss Function: Dice Loss + Cross-Entropy for class-imbalanced segmentation

Evaluation: Dice Score per sub-region, 3D slice-by-slice visualization

TECHNOLOGIES USED:

- PyTorch 2.x for model training & inference
- MONAI for 3D Attention U-Net architecture & medical image transforms
- nibabel for loading NIfTI (.nii.gz) BraTS 2021 volumes
- NumPy for 3D array manipulation & mask processing
- Matplotlib for 2D slice visualization & overlay plots
- Streamlit for interactive 3D demo application
- Kaggle Tesla P100-16GB for GPU-accelerated training

PROGRESSION MODULE

METHODOLOGY:

Longitudinal Tumor Growth Forecasting using Mathematical + Deep Learning hybrid

Mathematical Models: Exponential, Gompertz, Logistic, Linear growth curves

Deep Learning: Attention-LSTM time-series forecaster for residual correction

Hybrid Approach: $V_{\text{hybrid}}(t) = V_{\text{logistic}}(t) + \text{LSTM_correction}(\text{residual})$

3D U-Net Spatial Prediction: Predicts WHERE growth occurs (mask-to-mask)

Evaluation: MAE, R^2 , growth pattern fitting per tumor grade (HGG vs LGG)

TECHNOLOGIES USED:

- PyTorch 2.x for LSTM & 3D U-Net model training
- SciPy (curve_fit) for mathematical growth model parameter estimation
- Pandas for longitudinal patient data management & time-series formatting
- NumPy for volume extraction & numerical computation
- Plotly / Matplotlib for growth curve visualization & interactive dashboards
- Streamlit for interactive 3D progression demo application
- Dataset: MU-Glioma-Post (TCIA) — 203 patients, 654 predictions

IMPLEMENTATION: FEDQPSO

```
def aggregate_qpso(self, client_weights, client_val_accs):
    """
    QPSO Aggregation Step:
    1. Update personal best (pbest) and global best (gbest)
    2. Compute mean best (mbest)
    3. Apply quantum position update
    """

    # Step 1: Update pbest and gbest
    for client_id, weights in client_weights.items():
        val_acc = client_val_accs[client_id]

        if val_acc > self.personal_best_scores[client_id]:
            self.personal_best[client_id] = weights
            self.personal_best_scores[client_id] = val_acc

        if val_acc > self.global_best_score:
            self.global_best = weights
            self.global_best_score = val_acc

    # Step 2: Compute mean best (element-wise mean of all pbests)
    self.mean_best = compute_mean_state_dict(
        [self.personal_best[c_id] for c_id in self.clients]
    )

    perturbation = clip(perturbation, -0.1, 0.1) # Safety bounds

    new_position = p + random.choice([-1, 1]) * perturbation

    w_new[param_name] = new_position
```

```
    # Step 3: Quantum position update for each parameter
    for param_name in self.global_model.state_dict():
        # Per-parameter vectors
         $\varphi$  = random.uniform(0, 1) # per-client attraction weight
        u = random.uniform(0.3, 1.0) # quantum randomness

        # Attraction point: weighted combination of pbest and gbest
        p =  $\varphi$  * mean_best[param_name] + (1 -  $\varphi$ ) * gbest[param_name]

        # Quantum update: stochastic jump with bounded magnitude
        perturbation =  $\beta$  * abs(mean_best[param_name] - gbest[param_name]) * ln(1/u)

        # Attraction point: weighted combination of pbest and gbest
        p =  $\varphi$  * mean_best[param_name] + (1 -  $\varphi$ ) * gbest[param_name]

        # Quantum update: stochastic jump with bounded magnitude
        perturbation =  $\beta$  * abs(mean_best[param_name] - gbest[param_name]) * ln(1/u)
        perturbation = clip(perturbation, -0.1, 0.1) # Safety bounds

        new_position = p + random.choice([-1, 1]) * perturbation

        w_new[param_name] = new_position

    return w_new
```

IMPLEMENTATION: 3D ATTENTION U-NET SEGMENTATION

```
def segment_patient(model, patient_image_path, device):  
    """  
    Segment one patient's 3D MRI volume.  
    Args:  
        model: Trained AttentionUnet  
        patient_image_path: Path to .nii.gz file  
        device: torch device (cuda/cpu)  
    Returns:  
        predictions: (1, 3, H, W, D) segmentation mask  
    """  
  
    # Load and preprocess  
    sample = train_transforms({"image": patient_image_path,  
                              "label": patient_image_path})  
    image = sample["image"].unsqueeze(0).to(device) # (1, 4, H, W, D)  
  
    model.eval()  
    with torch.no_grad():  
        # Sliding window inference: process in 963 patches  
        outputs = sliding_window_inference(  
            inputs=image,  
            roi_size=(96, 96, 96),  
            sw_batch_size=4,  
            predictor=model,  
            overlap=0.5 # 50% overlap between patches  
        )  
  
        # Apply sigmoid + threshold  
        predictions = (outputs.sigmoid() > 0.5).float()  
  
    return predictions # (1, 3, H, W, D)
```

```
# Example usage  
model = AttentionUnet(spatial_dims=3, in_channels=4, out_channels=3)  
model.load_state_dict(torch.load("best_model.pth", map_location=device))  
  
pred = segment_patient(model, "patient_T1.nii.gz", device)  
print(f"Segmentation shape: {pred.shape}")  
print(f"Tumor voxels: TC={pred[0,0].sum():.0f}, "  
      f"WT={pred[0,1].sum():.0f}, ET={pred[0,2].sum():.0f}")
```

IMPLEMENTATION: TUMOUR PROGRESSION ANALYSIS

```
def train_hybrid_lstm(model, train_loader, val_loader, device,  
epochs=50):
```

```
    """Train LSTM on residual sequences"""
```

```
    optimizer = torch.optim.Adam(model.parameters(), lr=0.001)
```

```
    criterion = nn.MSELoss()
```

```
    for epoch in range(epochs):
```

```
        model.train()
```

```
        train_loss = 0.0
```

```
        for x_batch, y_batch in train_loader:
```

```
            x_batch = x_batch.to(device)
```

```
            y_batch = y_batch.to(device)
```

```
            optimizer.zero_grad()
```

```
            y_pred = model(x_batch)
```

```
            loss = criterion(y_pred, y_batch)
```

```
            loss.backward()
```

```
            optimizer.step()
```

```
            train_loss += loss.item()
```

```
    train_loss /= len(train_loader)
```

```
    # Validation
```

```
    model.eval()
```

```
    val_loss = 0.0
```

```
    with torch.no_grad():
```

```
        for x_batch, y_batch in val_loader:
```

```
            x_batch = x_batch.to(device)
```

```
            y_batch = y_batch.to(device)
```

```
            y_pred = model(x_batch)
```

```
            loss = criterion(y_pred, y_batch)
```

```
            val_loss += loss.item()
```

```
    val_loss /= len(val_loader)
```

```
    if epoch % 10 == 0:
```

```
        print(f"Epoch {epoch}: Train Loss={train_loss:.6f}, Val Loss=  
{val_loss:.6f}")
```

RESULTS AND OUTPUT (UPLOAD FILES)

NeuroAI v1.0

1 Upload2 Classify3 Segment4 Prognosis

Ready

Drop a ZIP file containing NIfTI modalities (T1, T1ce, T2, FLAIR) or upload them individually

Backend
FastAPI Inference API

Classification
QPSO-FL ResNet + BraTS Safeguard

Segmentation
3D Attention U-Net (MONAI)

Prognosis
Logistic + Spatial U-Net

Upload → Classify → Segment → Prognosis

System Status

☒ Classification

☒ Segmentation

☒ Spatial U-Net

↑

Drag & drop your files here

ZIP file with NIfTI modalities or individual .nii.gz files

Browse Files

RESULTS AND OUTPUT (CLASSIFICATION)

NeuroAI v1.0

1 Upload

2 Classify

3 Segment

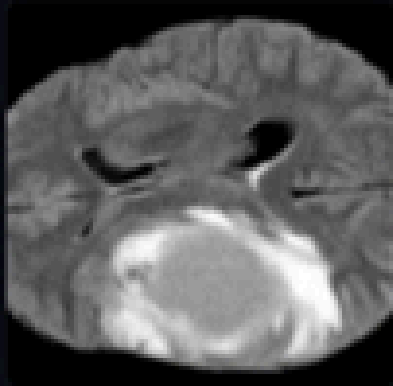
4 Prognosis

Classification complete

 **Tumor Classification**

QPSO-FL ResNet Inference with BraTS protocol safeguards

INPUT SLICE



FLAIR · Slice 74 / 155

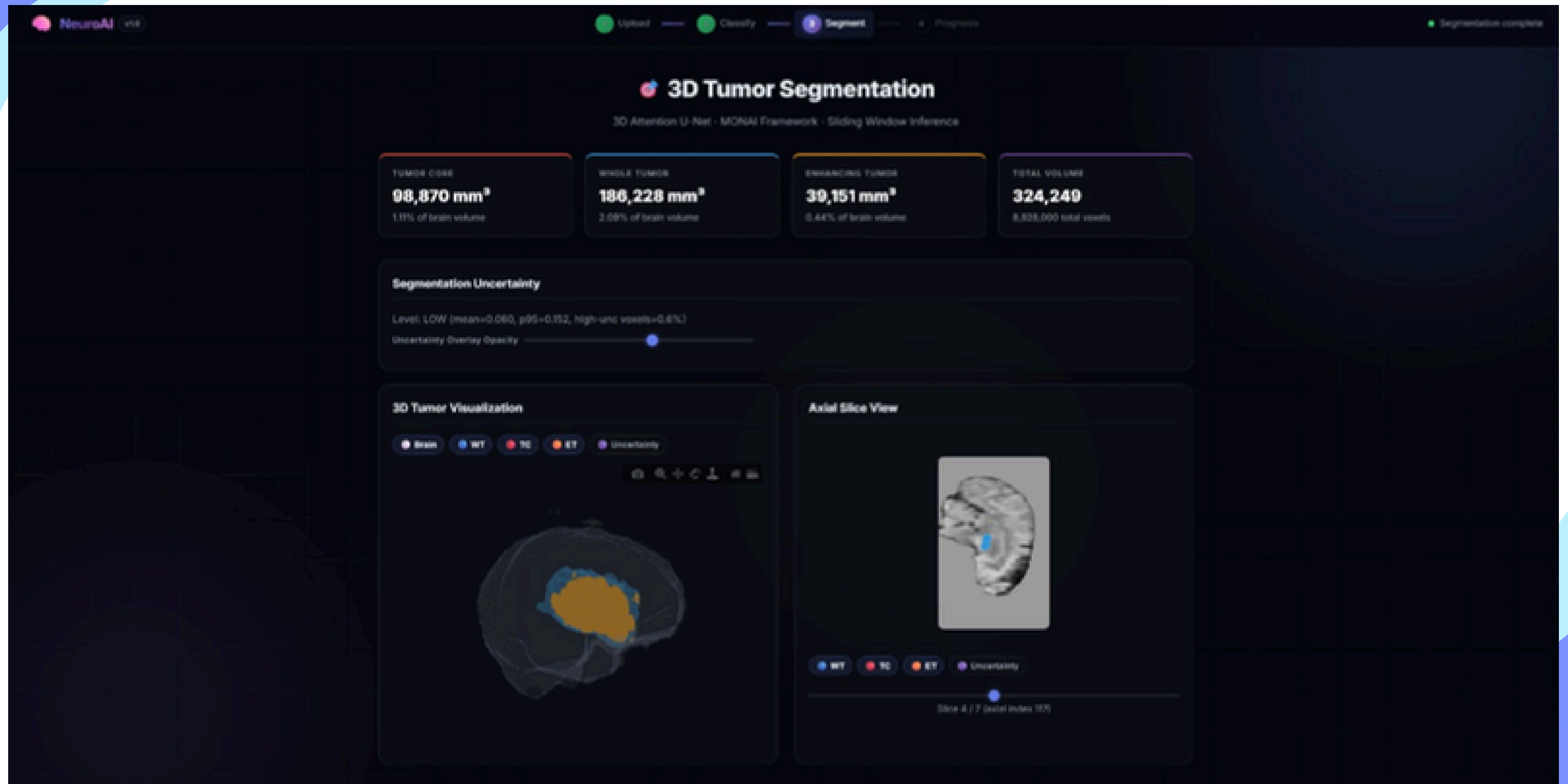
BraTS protocol detected: Dataset is glioma-only, so tumor-type classification was safely bypassed.



Glioma

Protocol-based clinical safeguard applied

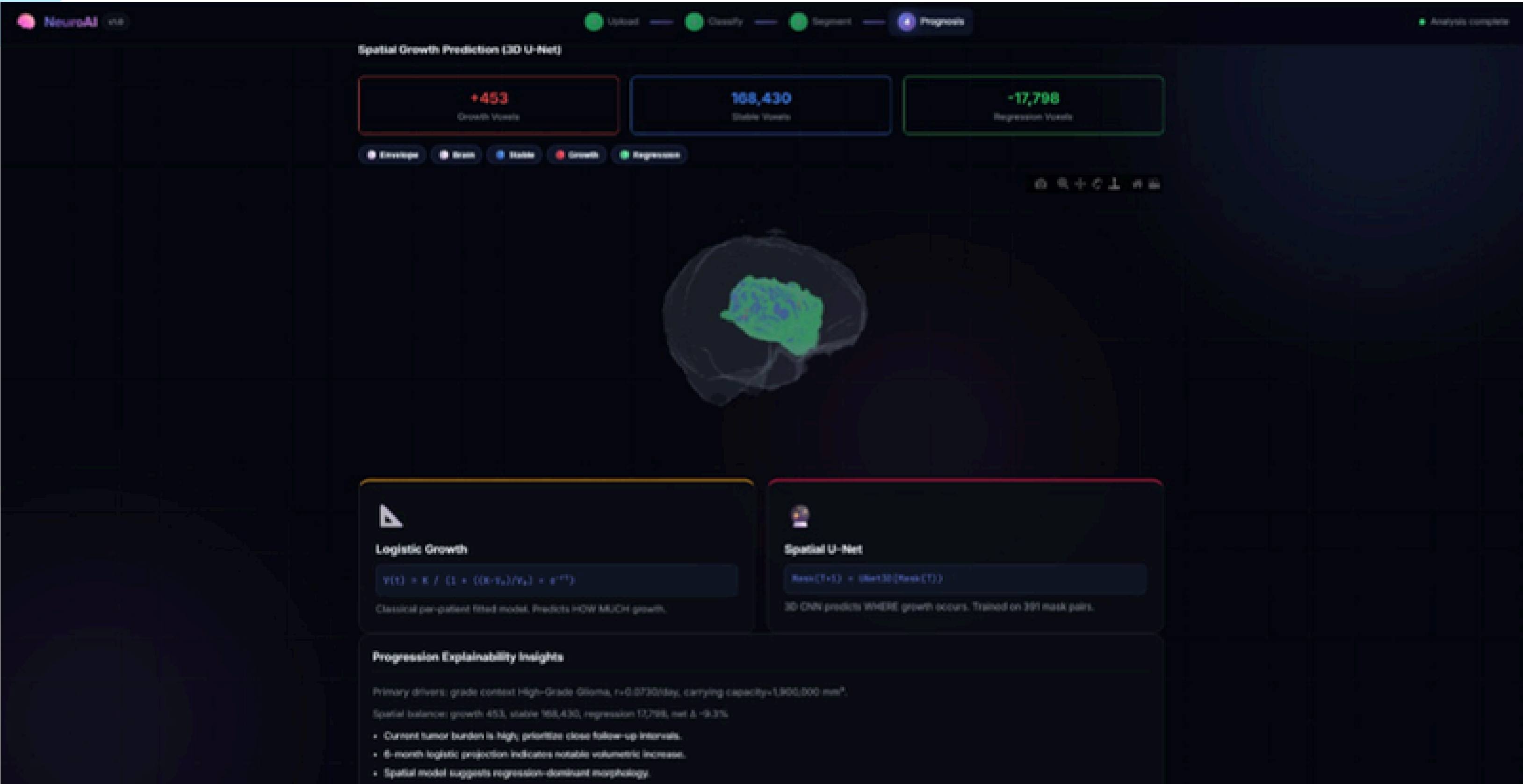
RESULTS AND OUTPUT (SEGMENTATION)



RESULTS AND OUTPUT (PROGRESSION)



RESULTS AND OUTPUT (PROGRESSION)



RESULT SCREEN

Metric	FedAvg	FedProx	L-b-L QPSO
Best Global Accuracy	90,56%	93,02%	92,09%
Accuracy Gap vs Best	-2,46%	—	-0,93%
Client 1 (smallest)	60,42%	77,50%	80,00%
Client 2	85,73%	86,80%	87,33%
Client 3 (largest)	89,17%	92,14%	88,10%
Client Std Dev	12,82%	6,05%	3,65%
Fairness Improvement	—	2.1x fairer than FedAvg	3.5x fairer than FedAvg

EXISTING VS PROPOSED (SEGMENTATION)

Aspect	Existing Methods (U-Net / Basic CNN)	Proposed Model (Your Enhanced Approach)
Segmentation Accuracy	Moderate accuracy; struggles with irregular tumor boundaries	Higher accuracy with better boundary detection using advanced feature extraction
Multi-class Segmentation	Limited (often binary segmentation: tumor vs non-tumor)	Supports multi-class segmentation (core, edema, enhancing tumor)
Generalization	Overfits on training data; poor performance on unseen scans	Better generalization due to improved training strategy / augmentation
Computational Efficiency	Faster but less optimized for complex cases	Slightly higher computation but optimized for performance-quality balance
Feature Extraction	Limited hierarchical feature capture	Deep feature extraction with attention / hybrid modules
Clinical Relevance	Less reliable for real-world diagnosis	More clinically useful with accurate size, shape, and location estimation

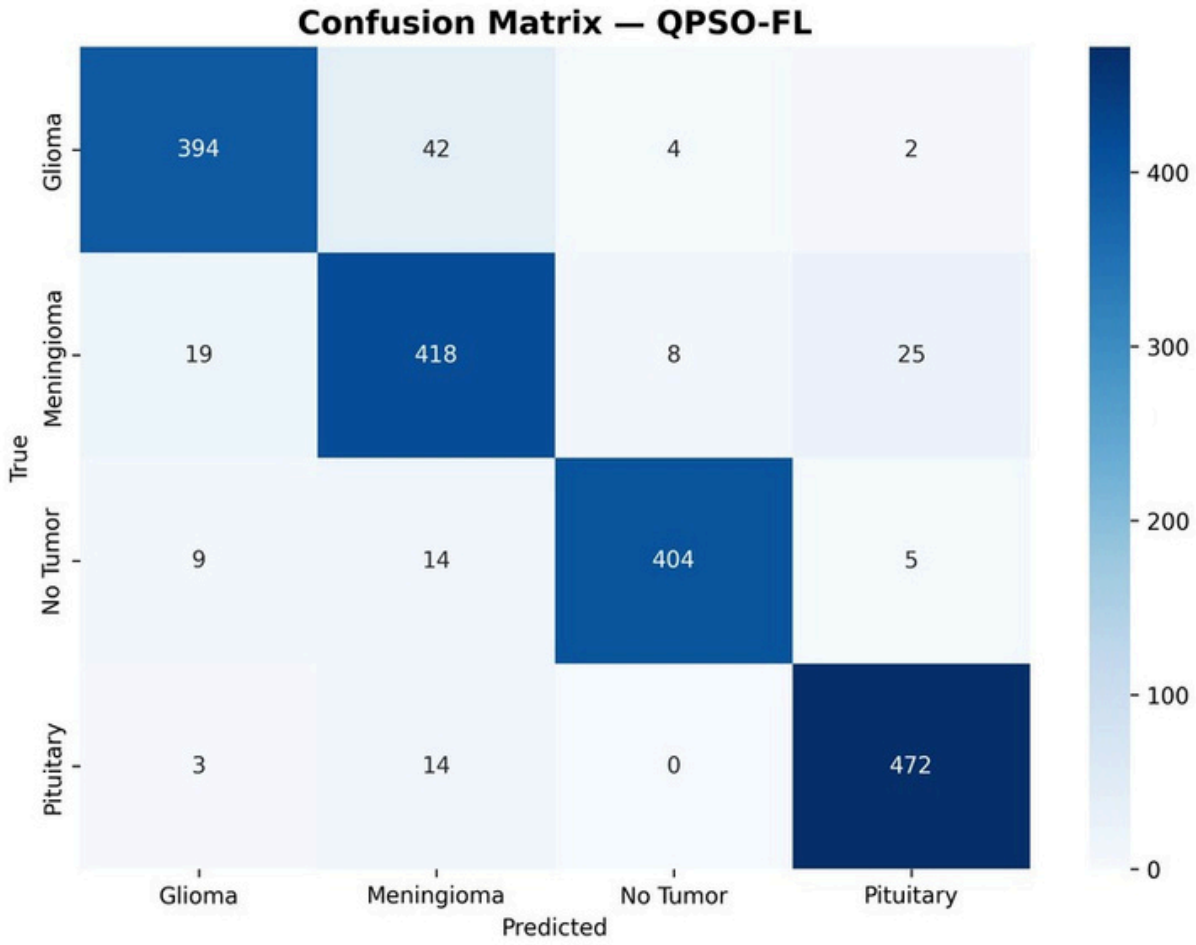
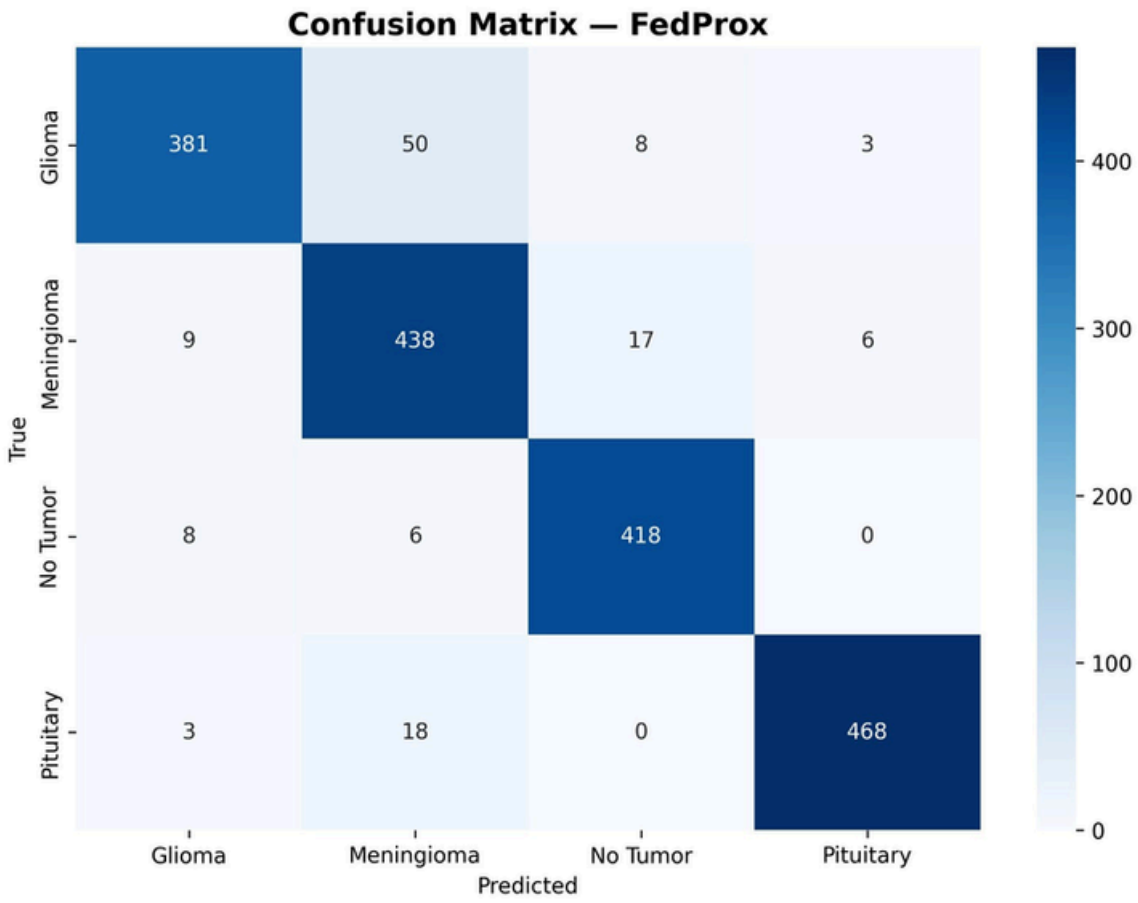
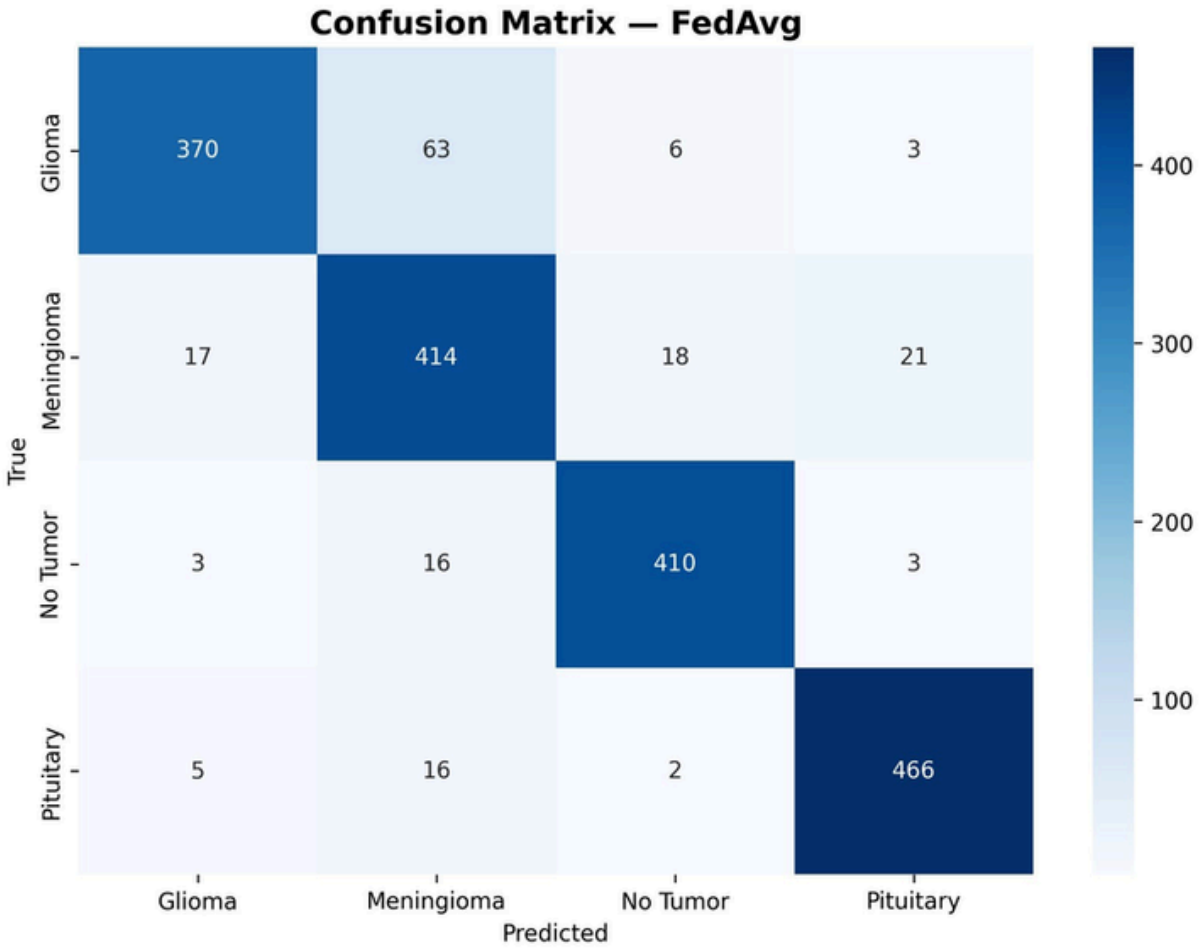
EXISTING VS. PROPOSED TUMOUR PROGRESSION FORECASTING

Feature	Existing Clinical Systems (RANO/RECIST)	Our Proposed System (Math + LSTM Hybrid)
Measurement Dimensionality	2D linear measures (perpendicular diameters)	3D total tumor volume (mm ³) extracted from segmentation masks
Temporal Perspective	Static/Retrospective: Only compares the current scan to the most recent baseline.	Longitudinal: scan-to-scan prediction using fitted growth trajectory
Kinetics Modeling	None: Growth is treated as a linear "jump" between two static points.	Biologically-Aware Models: Competitively fits 4 models (Exponential, Gompertz, Logistic, etc.).
Forecasting Engine	N/A: No predictive capability; purely reactive based on historical evidence.	Hybrid Intelligence: Logistic baseline + Attention-LSTM residual correction (grade-stratified HGG/LGG)
Decision Logic	Fixed Thresholds: Hard-coded % changes (e.g., PD = >25% increase).	Adaptive Model Selection: Per-patient logistic curve fitting with R ² goodness-of-fit evaluation
Grade Stratification	Not Addressed: Not differentiated by tumor grade	Separate HGG and LGG models with grade-specific growth rate parameters
Spatial Awareness	Minimal: Treats the tumor as a single diameter.	3D U-Net spatial prediction: predicts WHERE growth/regression occurs (mask → mask)

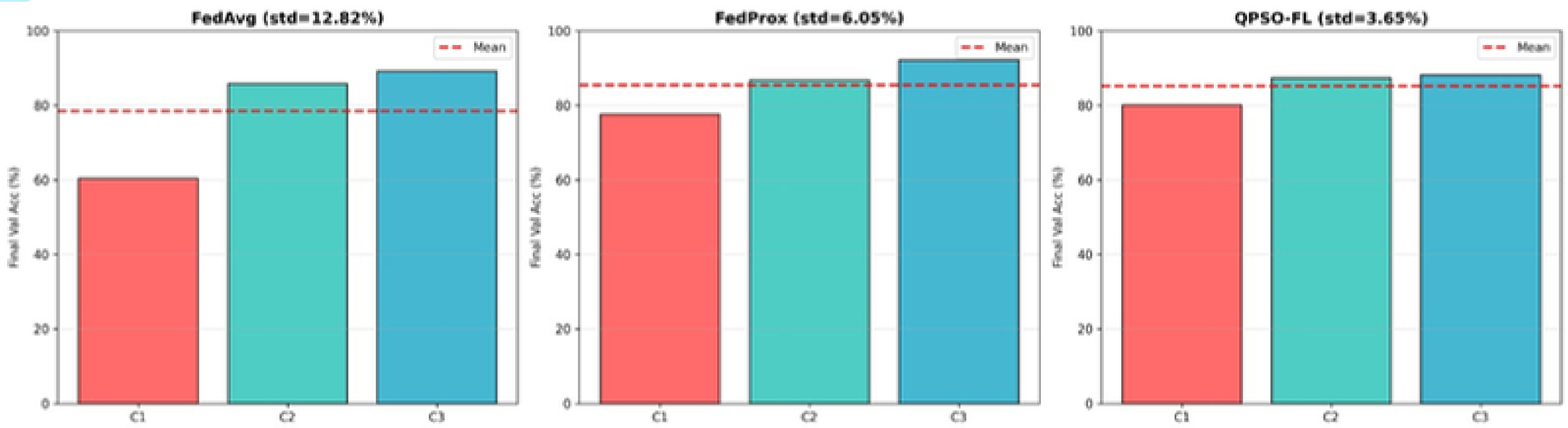
COMPARITIVE ANALYSIS

CLASSIC FL-ALGORITHMS VS FEDQPSO (PROPOSED TECHNIQUE)

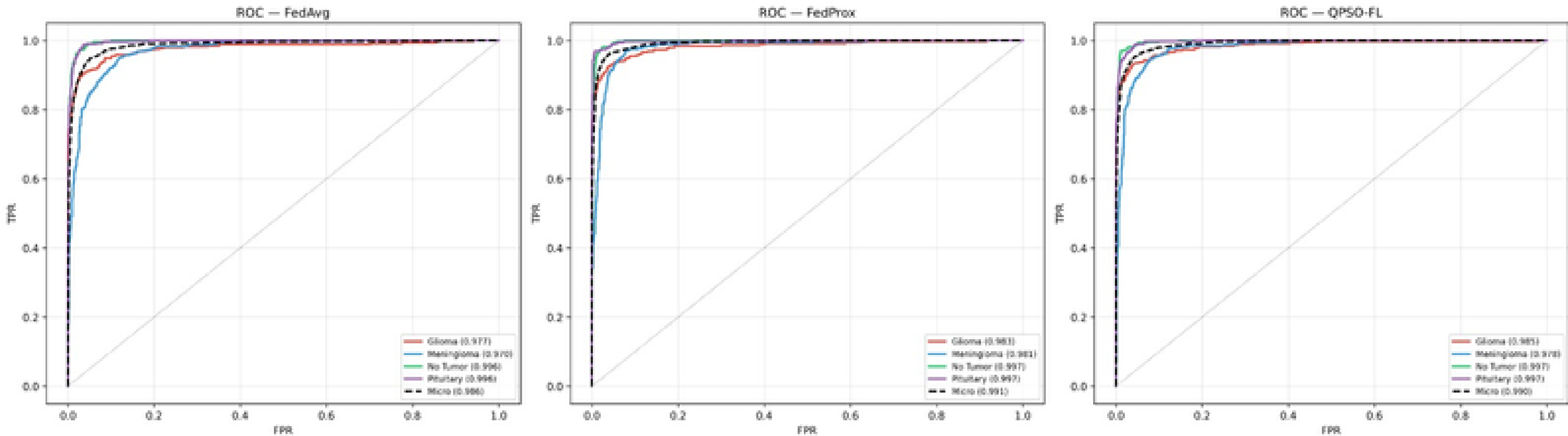
FOR BRAIN TUMOUR CLASSIFCATION.



EXISTING VS PROPOSED (CLASSIFICATION)



EXISTING VS PROPOSED (CLASSIFICATION)



CONCLUSION

Developed an end-to-end brain tumor management pipeline integrating classification, segmentation, and progression forecasting

Proposed FedQPSO — a novel layer-wise Quantum PSO aggregation algorithm that achieves implicit fairness without explicit regularizers

FedQPSO reduces inter-hospital accuracy gap by 81% vs FedAvg and is the only method keeping the weakest hospital $\geq 80\%$ under label skew

3D Attention U-Net achieves Mean Dice 0.76 on BraTS 2021 segmentation

Hybrid Math + LSTM progression model improves MAE by 7.88% for HGG patients

KEY TAKEAWAYS:

Global accuracy hides inequity — per-client fairness metrics are essential in clinical FL

FedQPSO's advantage scales with heterogeneity (Cohen's d : 0.35 $\rightarrow$ 1.26)

Highest Glioma recall in both setups — the most clinically critical class

PROJECT OUTCOME SCREENSHOT



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News

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Multi-language ▼

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FUTURE SCOPE

Short-Term (6–12 Months)

Differential Privacy: Implement DP-FedAvg & DP-QPSO-FL ($\epsilon = 1-10$); measure accuracy-privacy trade-off
Hyperparameter Tuning: Ablation studies on QPSO parameters; Bayesian/AutoML optimization
Hospital Pilot (3–5 Sites): Deploy FL server, collect prospective data, assess adoption barriers

Medium-Term (1–2 Years)

Federated Segmentation & Progression: Extend FL to segmentation (Module 2) & progression (Module 3) across 10+ hospitals
Personalization: Federated multi-task learning with per-hospital task heads & domain shift monitoring
Communication Efficiency: Gradient compression targeting 50% bandwidth reduction ($\leq 1\%$ accuracy loss)
Survival Forecasting: Federated Cox regression / deep survival models for OS & PFS prediction

Long-Term (2–5 Years)

Multi-Modal Fusion: Integrate genomics, clinical metadata, and radiomics with MRI features
Clinical Trials: Use QPSO-FL fairness metrics for patient stratification and publish prospective results
Foundation Models: Federated vision transformers on large-scale medical imaging with open-source release
FDA 510(k) Approval: Complete clinical validation for Class II medical device
Production Deployment: Scale to 20+ hospitals with real-world drift monitoring and federated governance

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Q & A



THANK
YOU!